Example 4.46 If ${}^nC_4 = 495$, What is n?.

Solution:
We know that,

$$ {}^nC_4 = 495. $$

Therefore,

$$ \frac{n \times (n-1) \times (n-2) \times (n-3)}{4 \times 3 \times 2 \times 1} = 495 $$

$$ \implies n \times (n-1) \times (n-2) \times (n-3) = 495 \times 4 \times 3 \times 2 \times 1 $$

Factoring $495 = 3 \times 3 \times 5 \times 11$, and writing this product as a product of 4 consecutive numbers in the descending order we get, $n \times (n-1) \times (n-2) \times (n-3) = 12 \times 11 \times 10 \times 9$. Equating n with the maximum number, we obtain $n = 12$.

Example 4.47 If ${}^nP_r = 11880$ and ${}^nC_r = 495$, Find n and r.

Solution:
We know that,

$$ \frac{{}^nP_r}{{}^nC_r} = r!. $$

Therefore,

$$ r! = \frac{11880}{495} = 24 = 4!, $$

gives $r = 4$. Using this $r = 4$, in ${}^nc_4 = 495$, and applying the result of the Example (4.46) we get, $n = 12$.

Example 4.48 Prove that ${}^{24}C_4 + \sum_{r=0}^{4} {}^{(28-r)}C_3 = {}^{29}C_4$

Solution:

$$ \begin{aligned} {}^{24}C_4 + \sum_{r=0}^{4} {}^{(28-r)}C_3 &= {}^{24}C_4 + {}^{28}C_3 + {}^{27}C_3 + {}^{26}C_3 + {}^{25}C_3 + {}^{24}C_3 \\ &= {}^{24}C_4 + {}^{24}C_3 + {}^{25}C_3 + {}^{26}C_3 + {}^{27}C_3 + {}^{28}C_3 \\ &= {}^{25}C_4 + {}^{25}C_3 + {}^{26}C_3 + {}^{27}C_3 + {}^{28}C_3 \\ &= {}^{26}C_4 + {}^{26}C_3 + {}^{27}C_3 + {}^{28}C_3 \\ &= {}^{27}C_4 + {}^{27}C_3 + {}^{28}C_3 \\ &= {}^{28}C_4 + {}^{28}C_3 \\ &= {}^{29}C_4 \end{aligned} $$